

Task 2: Product & Category Management - Documentation

1. Category & Product Table Design

Category Table:

- **Category_ID:** INT (Primary Key) - Uniquely identifies each category.
- **Category_Name:** VARCHAR(100) (NOT NULL, UNIQUE) - Name of the category.
- **Description:** TEXT - Brief summary of the category.

Product Table:

- **Product_ID:** INT (Primary Key) - Uniquely identifies each product.
- **Product_Name:** VARCHAR(150) (NOT NULL) - Name of the product.
- **Category_ID:** INT (Foreign Key referencing Category) - Links product to its category.
- **Price:** DECIMAL(10, 2) (NOT NULL, CHECK Price > 0) - Cost of the product.
- **Stock_Quantity:** INT (NOT NULL, CHECK Stock_Quantity >= 0) - Available inventory count.

2. ER Diagram Representation

Category (1) -----< (M) Product

- Category has fields: Category_ID (PK), Category_Name, Description
- Product has fields: Product_ID (PK), Product_Name, Category_ID (FK), Price, Stock_Quantity

3. Sample Product Data

ID	Product Name	Category	Price	Stock
101	Smartphone X	Electronics	\$699.99	50
102	Wireless Earbuds	Electronics	\$89.99	120
103	Gaming Laptop	Electronics	\$1,299.50	25
104	Cotton T-Shirt	Clothing	\$19.99	200
105	Denim Jeans	Clothing	\$49.99	80
106	Running Shoes	Clothing	\$79.99	60
107	Blender Pro	Home & Kitchen	\$45.00	40
108	Coffee Maker	Home & Kitchen	\$89.00	30
109	Non-Stick Pan Set	Home & Kitchen	\$65.50	45
110	Python Guide	Books	\$39.99	100
111	Sci-Fi Anthology	Books	\$24.99	75
112	Smart TV 55 inch	Electronics	\$450.00	15

4. Category-wise Analysis Report

- **Electronics:** Highest-priced product (Gaming Laptop at \$1,299.50) with the highest concentration of total value.
- **Clothing & Home/Kitchen:** Stable mid-range inventory items with consistent daily demand and pricing between \$20 and \$90.
- **Books:** Lower price points with strong stock counts supporting educational needs.
- **Average Price:** Across all 12 products, the average product price is approximately \$245.92.

5. Query Output & Verification

All SQL scripts including Database creation, Constraints enforcement, CRUD operations (INSERT, SELECT, UPDATE, DELETE), and Category-wise aggregate queries (COUNT, MAX, AVG) have been successfully verified and executed.